ORGAN TRANSPLANTATION

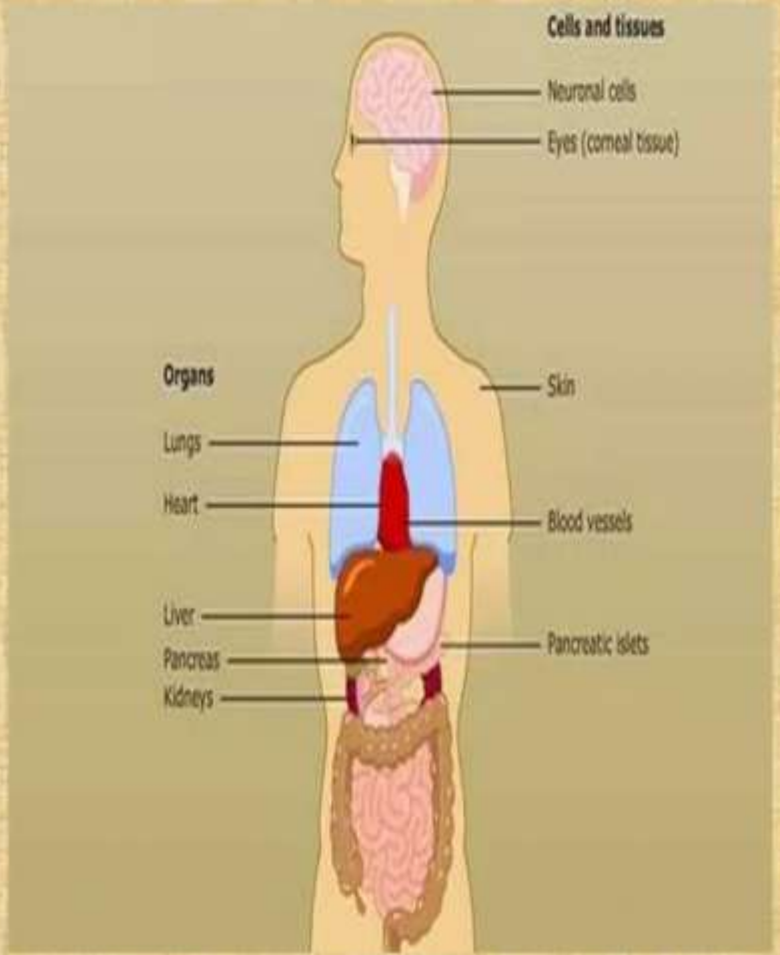
DR\Abdelrahman Elzefzaf
Consultant in Intensive Care and Critical Care
At Shebin El-Kom Specialized Surgery Hospital and Ismailia Medical Complex
Master's Degree in Critical Care Egyptian Fellowship in Intensive Care

INTRODUCTION

- Organ transplantation is the moving of a whole or partial organ from one body to another. (for the purpose of replacing the recipient's damaged or failing organ with a healthy , working organ)
- Organ transplants can be categorized as "*life saving*", while tissue transplants are "*life enhancing*".
- **Donor** : Individual who provides the organ
- **Recipient** : Individual who receives the organ

TRANSPLANTABLE ORGANS

- Liver
- Kidney
- Cornea
- Pancreas
- Lung
- Heart
- Intestine
- Face
- Bone marrow



TYPES OF DONOR

- **LIVING** :The donor remains alive and donates a renewable organ or part of an organ in which the remaining organ can regenerate.

  Example:- Single kidney donation, partial donation of liver, etc

- **DECEASED**: Donors (formerly cadaveric) are people who have been declared brain-dead and whose organs are kept viable by ventilators.

  Example:- Heart , stomach etc

TYPES OF TRANSPLANTATION

- Autograft
- Isograft
- Allograft
- Xenograft
- Split transplant
- Domino transplant

Figure ………………..→

A black leg is transplanted onto a white person.

TYPES OF TRANSPLANTATION

- **AUTOGRAFT**:- Autograft is the transfer of tissue/organ from one body site to another in the same individual.
- **ISOGRAFT**:- Isograft is the transfer of tissue/organ between genetically identical individuals.
- **ALLOGRAFT** :- An Allograft is the transfer of organ/tissue between genetically different individuals of the same species.

TYPES OF TRANSPLANTATION

- **XENOGRAFT**:- It is the transplant of organ/tissue from one species to another.
- **SPLIT TRANSPLANTATION** :- An organ of the deceased donor (specifically liver) may be divided between two recipients. (For example, between an adult and a child)
- **DOMINO TRANSPLANTATION** :- In this type of transplantation, the complete set of organ is transplanted. (In cystic fibrosis both the lungs need to be replaced)

GRAFT REJECTION

- Components of immune system involved in graft rejection:-
  1. **Antigen Presenting Cells** –
     Dendritic cells ,Macrophages , activated B - cells
  2. B - Cells and Antibodies
  3. T – Cells
  4. **Other cells** –
     Natural killer cells and Monocytes

TYPES OF GRAFT REJECTIONS

- **HYPER-ACUTE REJECTION** :- Begins within minutes or hours after hosts blood vessels are anatomised to graft vessels.
- **ACUTE REJECTION** :- Begins after the first week of transplantation.
- **CHRONIC REJECTION** :- Occurs in solid organs like heart , lungs. This rejection occurs with the loss of graft function over a prolonged period.

Chronic rejection

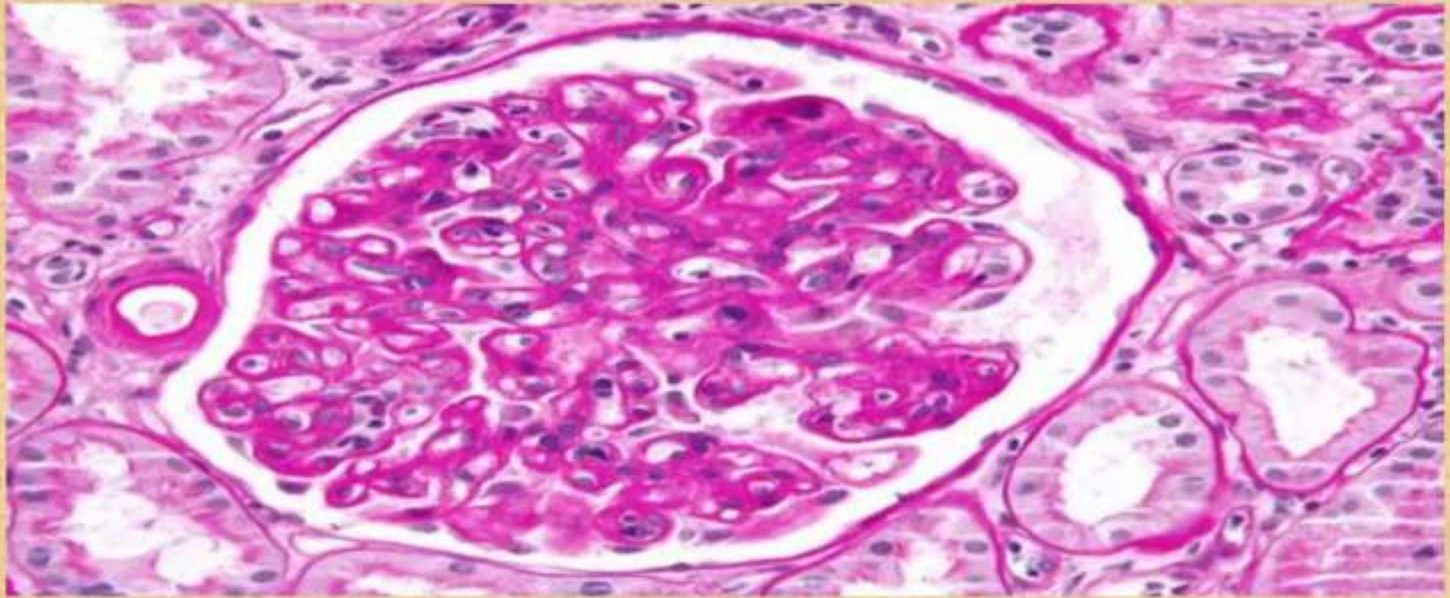


Figure :- Micrograph showing an abnormal glomerulus

- Loss in the function of the transplanted organ .
- Usually caused after 4-5 years of transplant.

REJECTION TREATMENT

- Hyper-acute and Acute rejections are treated by removal of tissue/organ.
- Chronic rejection is considered irreversible and can be treated only by **Retransplantation**.
- **IMMUNOSUPPRESSIVE THERAPY** –

  Immunosuppressive drugs are given as high doses for a short period of time.

  Example:- a. Coticosteriods (like Hydrocortisone)

  b. Anti – proliferatives (like Mycophenolic acid)

  c. Calcineurin inhibitors (like cyclosporin)

SUCCESSFUL TRANSPLANTATIONS

- 1905: First successful cornea transplant
- 1950: First successful kidney transplant
- 1967: First successful heart transplant
- 2008: First baby born from transplanted ovary
- 2013: First successful entire face transplantation

Thank You